VAN DER WAERDEN'S INEQUALITY BY MONOTONE AVERAGING

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A real $n \times n$ matrix $A = (a_{ij})$ is doubly stochastic if its entries are nonnegative and every row and column sums to one. Van der Waerden's (vdW) inequality states that every such matrix satisfies

$$\operatorname{per} A \geq n!/n^n, \qquad \text{with equality exactly when } A = U := \mathbf{1}\mathbf{1}^T/n.$$

Here $\operatorname{per} A = \sum_{\sigma \in \mathfrak{S}_n} \prod_i a_{i,\sigma(i)}$ is the permanent, $\mathfrak{S}_n$ is the symmetric group, and $\mathbf{1}$ is the all-ones vector. Falikman (1981) and Egorychev (1981) proved the inequality using the quadratic Alexandrov–Fenchel inequality. Gurvits (2008) later gave a proof via stable polynomials. We give a new, short, and self-contained proof by monotone averaging of a different quantity.

The monotone quantity measures the dependence in a random permutation. For a doubly stochastic P, let $\mathcal{M}(P)$ be the probability laws on $\mathfrak{S}_n$ with $\Pr(\sigma(i) = j) = p_{ij}$, and set

$$H(P) = \max_{\mu \in \mathcal{M}(P)} \mathrm{H}(\mu), \qquad S(P) = -\sum\nolimits_{ij} p_{ij} \log p_{ij}, \qquad I(P) = S(P) - H(P).$$

Here $\mathrm{H}(\mu) = -\sum_\sigma \mu(\sigma) \log \mu(\sigma)$. Thus $I(P)$ is the minimum total correlation among permutation laws with marginals P. We prove that I never decreases when two rows of P are averaged (Theorem 4), so I is largest at U, which finite averages reach; a relative-entropy comparison then gives vdW. The derivative along an average is nonnegative because the cofactors of the two averaged rows satisfy the quadratic Alexandrov–Fenchel inequality.

Lemma 1 (Entropy differential)**.** *For positive doubly stochastic P, its entropy-maximizing law has the form $\mu_B(\sigma) = \prod_i b_{i,\sigma(i)} / \operatorname{per} B$, where $B > 0$ can be chosen smoothly in P, with $B(U) = \mathbf{1}\mathbf{1}^T$. For $\mathcal{V} = \{X \in \mathbb{R}^{n \times n} : X\mathbf{1} = X^T\mathbf{1} = 0\}$,*

$$DI(P)[X] = \sum\nolimits_{ij} X_{ij} \log(b_{ij}/p_{ij}), \qquad X \in \mathcal{V}. \tag{1}$$

Proof. By Birkhoff's theorem and compactness, $\mathcal{M}(P)$ contains an entropy maximizer, unique by strict concavity. For $P > 0$, writing $P = \delta U + (1 - \delta)R$ with $\delta > 0$ and R doubly stochastic, a mixture of the uniform law and a law for R is a positive law in $\mathcal{M}(P)$. Since entropy has infinite slope at zero masses, the maximizer is then positive, and, as in Brégman (1973, Theorem 2 and equation (9)), Lagrange multipliers give $b_{ij} = e^{\theta_{ij}}$.

The function $F(\theta) = \log \operatorname{per}(e^{\theta_{ij}})$, with entrywise exponentiation, has gradient P, and its Hessian is the covariance of the permutation matrix. This covariance is positive definite on $\mathcal{V}$, because a statistic $\sum_i X_{i,\sigma(i)}$ that is constant in σ forces X to be a sum of row and column potentials, which is zero on $\mathcal{V}$. Restricting θ to $\mathcal{V}$, the inverse function theorem gives a smooth choice of B with $B(U) = \mathbf{1}\mathbf{1}^T$, and $H(P) = F(\theta) - \langle P, \theta \rangle$ gives $DH(P)[X] = -\langle \theta, X \rangle$, which proves (1). □

Along an average of two rows, (1) becomes the pairing of Lemma 2 for a stochastic T built from the cofactors (see (2)), which two Jensen inequalities make nonnegative; quadratic information also bounds entropy gaps in Cryan et al. (2021, proof of Lemma 10).

Lemma 2 (Kernel inequality)**.** *Let T be reversible stochastic with stationary probability vector $\pi > 0$, and write $\langle u, v \rangle_\pi = \sum_j \pi_j u_j v_j$. If $\langle z, Tz \rangle_\pi \leq \langle z, \mathbf{1} \rangle_\pi^2$ for every real z, then $\langle Tz - z, \log(Tz) \rangle_\pi \geq 0$ for $z > 0$, with equality exactly when Tz is constant.*

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Proof. Set $a = \langle z, \mathbf{1}\rangle_\pi$. Jensen with weights $\pi_j z_j/a$, then Jensen for $s\log s$, gives

$$\langle z, \log(Tz)\rangle_\pi \le a \log \frac{\langle z, Tz\rangle_\pi}{a} \le a\log a \le \langle Tz, \log(Tz)\rangle_\pi.$$

Stationarity gives $\langle Tz, \mathbf{1}\rangle_\pi = a$, so the last inequality is strict unless $Tz = a\mathbf{1}$. □

We now require the following hyperbolic inequality (form of Alexandrov–Fenchel).

Lemma 3 (Cofactor inequality)**.** *Let $n \ge 2$ and fix $n-2$ positive rows in $\mathbb{R}^n$. Let $M_{jj} = 0$, and for $j \ne k$ let M_{jk} be their permanent after deleting columns j, k. For $n = 2$, set $M_{12} = M_{21} = 1$. Then $(x^T My)^2 \ge (x^T Mx)(y^T My)$ for $x \in \mathbb{R}^n$ and $y > 0$.*

Proof. A real polynomial is stable if it has no zero with all variables in the upper half-plane. If q is stable, homogeneous of positive degree with nonnegative coefficients, and $u > 0$, then for $\operatorname{Im} z > 0$ the polynomial $q(z+tu)$ in t has leading coefficient $q(u) > 0$ and roots α with $\operatorname{Im}\alpha < 0$, so $D_u q(z)/q(z) = \sum_\alpha (-\alpha)^{-1}$ has negative imaginary part and $D_u q$ is stable. Differentiating $z_1 \cdots z_n$ in the fixed row directions gives the stable quadratic $q(z) = \frac{1}{2} z^T Mz$. For real x and $y > 0$, stability and conjugacy make the roots of $q(x+ty)$ real; its nonnegative discriminant is the inequality. □

Theorem 4 (Monotone averaging)**.** *Let S_{rs} interchange rows r, s. For every doubly stochastic P, the function $t \mapsto I(((1-t)\mathrm{Id} + tS_{rs})P)$ is nondecreasing on $[0, 1/2]$. Consequently*

$$I(P) \le I(U) = n\log n - \log(n!), \qquad \text{with equality exactly when } P = U.$$

Proof. Let $n \ge 2$ ($n = 1$ is immediate) and first $P > 0$. We write x and y for rows r and s of B from Lemma 1, form M from its other rows as in Lemma 3, and set, dividing coordinatewise,

$$C = \operatorname{diag}(y) M \operatorname{diag}(y), \quad \rho = C\mathbf{1}, \quad T = \operatorname{diag}(\rho)^{-1} C, \quad z = x/y, \quad Z = \operatorname{per} B = \rho^T z.$$

Then T is reversible stochastic with stationary law $\pi = \rho/(\mathbf{1}^T\rho)$, and Lemma 3 gives $w^T Cw \le (w^T C\mathbf{1})^2/(\mathbf{1}^T C\mathbf{1})$ for all w, as Lemma 2 requires. Rows r and s of P, called u and v, satisfy

$$u_j = \rho_j z_j/Z, \qquad v_j = \rho_j (Tz)_j/Z, \qquad \log(x_j/u_j) - \log(y_j/v_j) = \log(Tz)_j.$$

In the direction changing only these two rows, (1) and Lemma 2 give

$$DI(P)[(v-u, u-v)] = \frac{1}{Z}\sum\nolimits_j \rho_j\big((Tz)_j - z_j\big)\log(Tz)_j \ge 0. \tag{2}$$

At path parameter $t < 1/2$, rows r and s differ by $(1-2t)(v-u)$, so the derivative of I along the path is (2) at the current matrix divided by $1-2t$, hence nonnegative. Continuity gives $t = 1/2$.

For boundary P we use $P_\varepsilon = (1-\varepsilon)P + \varepsilon U$. Concavity of H and limits of maximizing laws give $I(P_\varepsilon) \to I(P)$; since averaging fixes U, monotonicity passes to the limit.

By induction, pair averages make the first $n-1$ rows equal to their mean w; averaging the last row with each copy of w in turn, at parameters $1/n, 1/(n-1), \ldots, 1/2$, moves every row to the overall mean, $\mathbf{1}^T/n$ for doubly stochastic P. Hence $I(P) \le I(U) = n\log n - \log(n!)$, since $H(U) = \log(n!)$.

For uniqueness, at U the choice $B(U) = \mathbf{1}\mathbf{1}^T$ gives $T = (\mathbf{1}\mathbf{1}^T - \mathrm{Id})/(n-1)$, which is invertible, as is T near U. There, by Lemma 2, a zero derivative in (2) forces Tz, hence z, to be constant, and $u = v$. On the last nontrivial segment of a chain ending at U, the averaged rows are unequal before the endpoint, so (2) is positive near it and $I(P) < I(U)$ for $P \ne U$, including boundary P. □

Proof of the permanent bound and equality case. Let μ maximize entropy with marginals A, and let $\nu(\sigma) = \prod_i a_{i,\sigma(i)}/\operatorname{per} A$, which is positive wherever μ is, even when A has zero entries ($\operatorname{per} A > 0$ by Birkhoff's theorem). On the support of μ, Jensen's inequality gives

$$0 \le \sum\nolimits_\sigma \mu(\sigma)\log\frac{\mu(\sigma)}{\nu(\sigma)} = \log\operatorname{per} A + I(A), \qquad \log\operatorname{per} A \ge -I(A) \ge \log(n!) - n\log n.$$

Equality forces $A = U$ by Theorem 4; direct evaluation gives $\operatorname{per} U = n!/n^n$. □

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AI usage

We used GPT-6 Pro and Claude (Opus 5.5) extensively in the preparation of this note; the key prompt was to study Schur-type averaging for permanents; see the companion paper (Sra, 2026) for a fuller account.

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